

DP Computer Science: A 3.4.1 Different Types of Database (HL Only)

1. Core Concepts & Learning Objectives

Learning Intentions

By the end of this lesson, students will be able to:

- Outline the key characteristics of **NoSQL**, **cloud**, **spatial**, and **in-memory** database models
- Identify **real-world scenarios** where each of these database models might be effectively used
- Explain why different database models are suited to different **data storage needs**
- Begin to consider how data manipulation might vary conceptually across different database models

Key Vocabulary

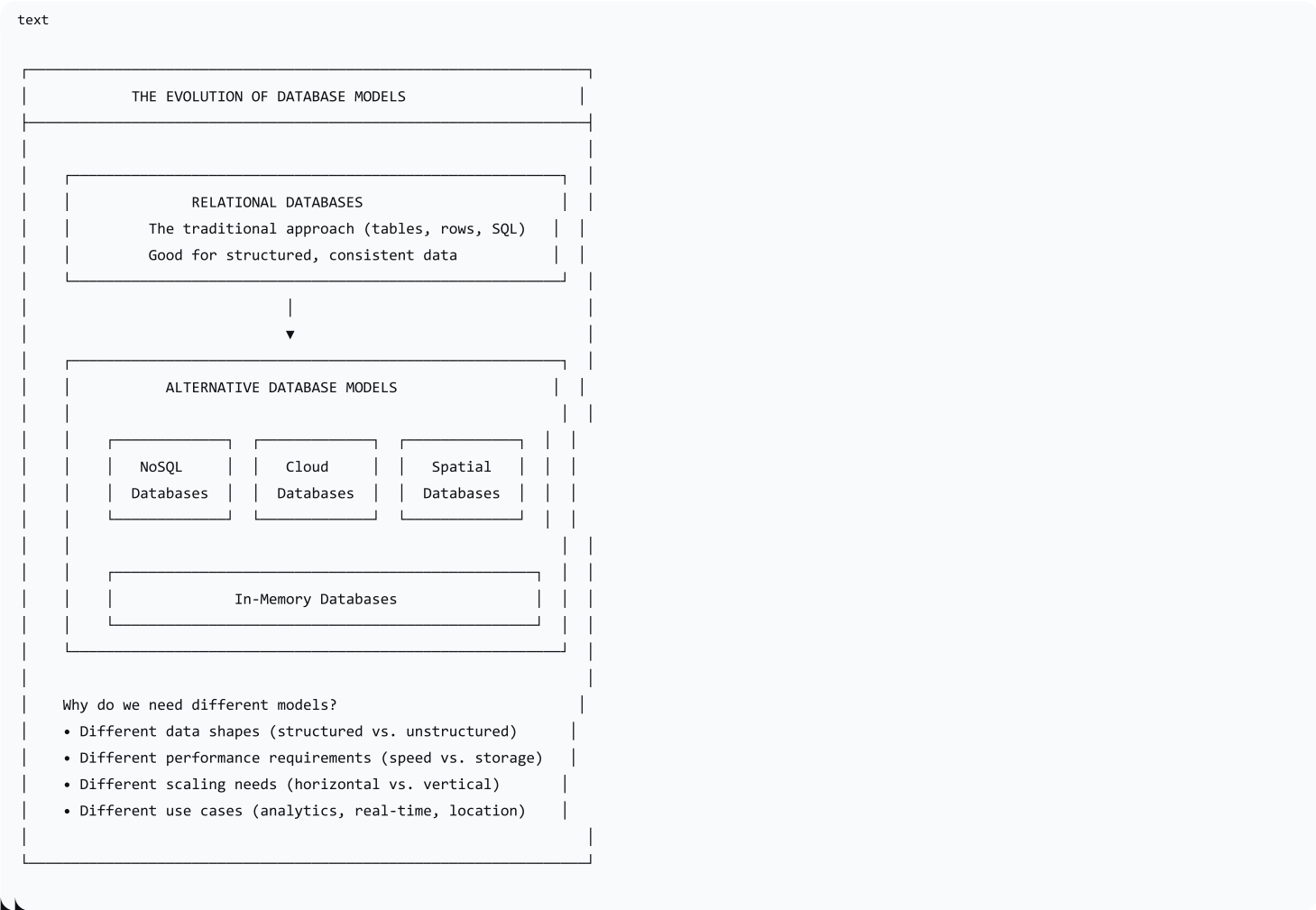
Term	Definition
Relational database	A database that organises data into tables with rows and columns, linked by relationships (e.g., SQLite, MySQL, PostgreSQL)
NoSQL database	A database that stores data in non-tabular formats (e.g., documents, key-value pairs, graphs, wide-column stores)
Cloud database	A database that is hosted, managed, and accessed via the cloud (e.g., AWS RDS, Google Cloud Spanner)
Spatial database	A database optimised for storing and querying geographical and location-based data (e.g., PostGIS)
In-memory database	A database that stores data primarily in RAM for ultra-fast access (e.g., Redis, Memcached)
Data warehouse	A specialised database for storing and analysing large volumes of historical data (HL extension)
Scalability	The ability of a database to handle increasing amounts of data or users
Horizontal scaling	Adding more servers to handle increased load (scaling out)
Vertical scaling	Adding more resources (CPU, RAM) to a single server (scaling up)

Relevant ATL Skills

Skill	Description
Thinking	Concept formation; transfer (applying to real-world examples); critical thinking
Communication	Presenting; listening during explanations and discussions
Research	Information literacy (brief research during extension activities)
Self-Management	Organisation (structuring notes and worksheet activities)

2. Introduction: Why Different Database Models?

The Evolution of Database Models



Comparison of Database Models at a Glance

Model	Data Structure	Speed	Scalability	Best For
Relational	Tables (rows/columns)	Good	Vertical	Structured data, transactions
NoSQL	Documents/Key-Value	Fast	Horizontal	Unstructured data, big data
Cloud	Managed services	Variable	Elastic	SaaS, global applications
Spatial	Geographic objects	Depends	Depends	GIS, location-based services
In-Memory	RAM-based	Ultra-fast	Limited	Real-time, caching, analytics

3. NoSQL Databases

What is a NoSQL Database?

NoSQL (Not Only SQL) databases store data in non-tabular formats such as documents, key-value pairs, graphs, or wide-column stores. They are designed for flexibility, scalability, and handling large volumes of unstructured or semi-structured data.

Key Characteristics

Characteristic	Description
Flexible Schema	No fixed table structure; fields can vary between records
Horizontal Scalability	Easy to add more servers to handle growth
Non-Relational	Does not require joins between tables
Varied Data Models	Document, key-value, graph, wide-column
High Performance	Optimised for specific use cases
Distributed	Often designed to run on clusters of servers

Types of NoSQL Databases

Type	Description	Example
Document Store	Stores data as documents (JSON, BSON)	MongoDB, CouchDB
Key-Value Store	Stores data as key-value pairs	Redis, DynamoDB
Graph Database	Stores data as nodes and relationships	Neo4j, Amazon Neptune
Wide-Column Store	Stores data in column families	Cassandra, HBase

Real-World Applications

Application	Why NoSQL?
Social Media	Handles diverse, unstructured user data (posts, comments, likes)
E-Commerce	Product catalogs with varying attributes
Content Management	Flexible content types and metadata
Real-Time Analytics	High-volume data ingestion and querying
IoT Data	Millions of devices sending different types of data

Analogy: A relational database is like a **spreadsheet**—everything must fit into rows and columns. A NoSQL document database is like a **folder of documents**—each document can have different fields and formats.

Example: MongoDB Document

```
json

{
  "user_id": "12345",
  "name": "Alice Smith",
  "email": "alice@example.com",
  "preferences": {
    "language": "en",
    "timezone": "UTC"
  },
  "purchase_history": [
    {"item": "Book", "price": 15.99},
    {"item": "Pen", "price": 2.50}
  ]
}
```

```
    ]
  }
```

4. Cloud Databases

What is a Cloud Database?

A cloud database is a database that is hosted, managed, and accessed via the cloud. It is offered as a service (DBaaS—Database as a Service) by cloud providers.

Key Characteristics

Characteristic	Description
Managed Service	The provider handles maintenance, backups, and updates
Elastic Scalability	Resources can be scaled up or down on demand
High Availability	Built-in redundancy and failover
Global Access	Accessible from anywhere with internet
Pay-as-you-go	Pay only for what you use
Integrated Services	Works with other cloud services (analytics, AI, etc.)

Real-World Applications

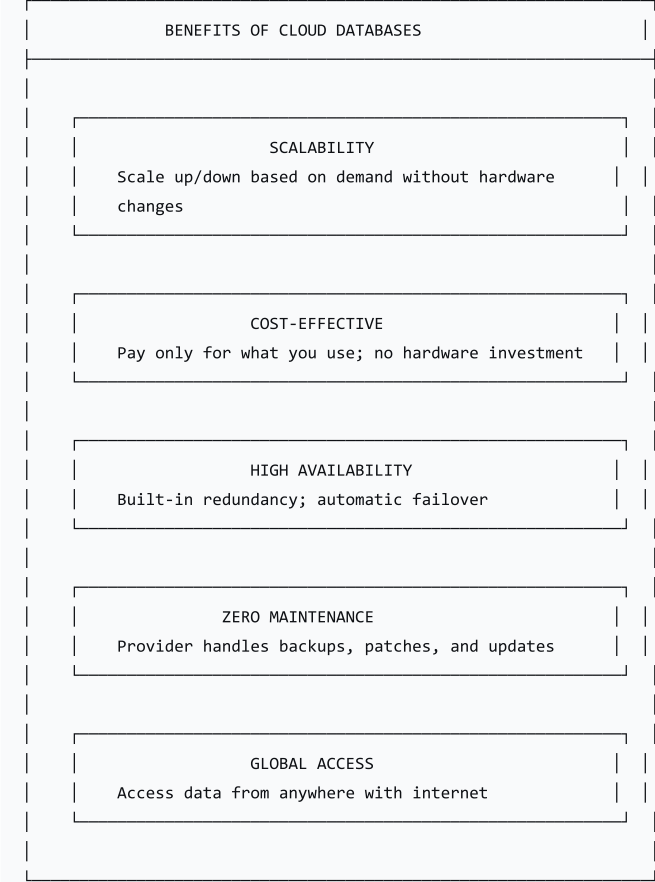
Application	Why Cloud Database?
SaaS Applications	Hosted applications need scalable, managed databases
Global Enterprises	Need data accessible across regions
Startups	No upfront hardware costs; pay-as-you-go
Disaster Recovery	Built-in backup and failover
Mobile Apps	Needs reliable, scalable backend storage

Cloud Database Examples

Provider	Service	Database Type
AWS	RDS, DynamoDB, Aurora	Relational, NoSQL
Google Cloud	Cloud SQL, Cloud Spanner	Relational, Distributed
Microsoft Azure	SQL Database, Cosmos DB	Relational, NoSQL
MongoDB Atlas	MongoDB as a Service	NoSQL

Benefits of Cloud Databases

```
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```



5. Spatial Databases

What is a Spatial Database?

A spatial database is optimised for storing and querying geographical and location-based data. It supports spatial data types (points, lines, polygons) and spatial queries (distance, proximity, containment).

Key Characteristics

Characteristic	Description
Geometric Data Types	Points, lines, polygons, multi-polygons
Spatial Indexing	Specialised indexes for fast location queries (R-tree, Quadtree)
Spatial Queries	Distance, intersection, containment, nearest-neighbour
Geographic Support	Latitude/longitude, projections, coordinate systems
Integration	Often used with GIS (Geographic Information Systems)

Real-World Applications

Application	Why Spatial Database?
GIS (Geographic Information Systems)	Mapping, city planning, environmental monitoring

Application	Why Spatial Database?
Location-Based Services	"Find nearby restaurants"
Logistics	Route planning, delivery optimisation
Public Safety	Emergency response planning
Real Estate	Property searches by location
Autonomous Vehicles	Navigation and obstacle detection

Spatial Database Example

PostGIS (PostgreSQL extension):

```
sql

-- Create a table with spatial data
CREATE TABLE locations (
  id INTEGER PRIMARY KEY,
  name TEXT,
  geom GEOMETRY(Point, 4326) -- Latitude/Longitude
);

-- Insert a point (New York)
INSERT INTO locations (name, geom)
VALUES ('New York', ST_GeomFromText('POINT(-74.006 40.7128)', 4326));

-- Find Locations within 10 km
SELECT name
FROM locations
WHERE ST_DWithin(geom, ST_GeomFromText('POINT(-74.006 40.7128)', 4326), 10000);
```

Analogy: A spatial database is like a **map**—it understands location, distance, and relationships between places.

6. In-Memory Databases

What is an In-Memory Database?

An in-memory database stores data primarily in RAM (Random Access Memory) rather than on disk. This provides ultra-fast data access for applications that require real-time performance.

Key Characteristics

Characteristic	Description
Ultra-Fast Access	Data is retrieved from RAM, not disk
Low Latency	Millisecond or microsecond response times
Volatile Storage	Data is lost when power is off (unless persisted)
Caching	Often used as a cache for slower disk-based databases
Limited Capacity	RAM is more expensive and limited than disk storage

Real-World Applications

Application	Why In-Memory Database?
Real-Time Analytics	Instant insights on streaming data
E-Commerce	Session management, real-time inventory
Gaming	Leaderboards, player states
Fraud Detection	Real-time pattern matching
Stock Trading	Microsecond-level transaction processing
Caching	Storing frequently accessed data (Redis, Memcached)

In-Memory Database Example

Redis (Key-Value Store):

```
python

# Set a value
SET user:1234:name "Alice Smith"

# Get a value
GET user:1234:name
# Returns: "Alice Smith"

# Store session data
SET session:abc123 "Logged in user"
EXPIRE session:abc123 3600 # Expires in 1 hour
```

Analogy: An in-memory database is like a **whiteboard**—writing and reading is fast, but if the board is erased (power loss), the data is gone.

7. Comparison Summary Table

Feature	Relational	NoSQL	Cloud	Spatial	In-Memory
Data Structure	Tables (rows/columns)	Documents/Key-Value	Varies	Geographic objects	Key-Value/Objects
Schema	Fixed	Flexible	Varies	Fixed/Flexible	Flexible
Scaling	Vertical	Horizontal	Elastic	Depends	Horizontal
Speed	Good	Fast	Variable	Depends	Ultra-fast
Storage	Disk	Disk	Disk (managed)	Disk	RAM
Durability	High	High	High (managed)	High	Low (volatile)
Best For	Structured data	Big data, unstructured	Managed services	GIS, location	Real-time, caching

8. Lesson Sequence

Lesson Plan (60 minutes)

Phase	Time	Activity
Introduction	5 min	Recap relational databases; introduce need for alternative models; state learning objectives
NoSQL Databases	10 min	Definition, key characteristics, real-world examples (Social Media). Class discussion: benefits of flexibility
Cloud Databases	10 min	Definition, key characteristics, real-world examples (SaaS, Managed Services). Question: How does the cloud address scalability?
Spatial Databases	10 min	Definition, key characteristics, real-world examples (GIS). Think-Pair-Share: Other applications using location data
In-Memory Databases	10 min	Definition, key characteristics, real-world examples (Real-Time Analytics, E-commerce). Why is speed critical?
Worksheet Activity	10 min	Students complete worksheet; consider how SQL data manipulation relates to different needs
Conclusion	5 min	Recap key differences; introduce future topics (Data Warehouses); Q&A

Discussion Questions

Question	Purpose
"Why would a social media platform use a NoSQL database instead of a relational database?"	Understand NoSQL benefits
"How does cloud computing address scalability challenges?"	Understand cloud databases
"What other applications might use a spatial database?"	Apply spatial database concepts
"Why is speed critical for in-memory databases?"	Understand in-memory purpose
"What is the trade-off between speed and durability in in-memory databases?"	Evaluate in-memory limitations

9. Worksheet Activity

Sample Worksheet Questions

Task	Description
Task 1	List four types of database models discussed in the lesson
Task 2	Match each database model to its best use case
Task 3	Explain why NoSQL databases are flexible

Task	Description
Task 4	Describe one application of a spatial database
Task 5	Discuss the trade-offs of using an in-memory database

10. Assessment Activities

Assessment Type	Description
Class Participation	Observe engagement and contributions
Worksheet	Evaluate understanding of different database models
Exit Ticket	Gauge understanding and identify areas needing clarification

Sample Exit Ticket Questions:

1. What is the main characteristic of a NoSQL database?
2. Why are cloud databases cost-effective?
3. What type of data is stored in a spatial database?
4. What is the main advantage of an in-memory database?

11. Differentiation

Level	Strategy
Support	Provide visual aids, analogies, and real-world examples; work in smaller groups
Challenge	Encourage research on additional examples; explore data warehouses; delve into technical differences between models

12. Resources

- **Presentation** (Slide Deck)
- **eBook** (A3.4.1 with examples)
- **Worksheet:** Database model matching and scenarios
- **Worksheet Answers**

13. Summary: Key Takeaways

Model	Key Feature	Best For
Relational	Structured tables, SQL, ACID	Traditional business applications
NoSQL	Flexible schema, horizontal scaling	Big data, social media, unstructured data
Cloud	Managed service, elastic scaling	SaaS, global applications, startups
Spatial	Geographic data, location queries	GIS, mapping, logistics

Model	Key Feature	Best For
In-Memory	Ultra-fast (RAM), low latency	Real-time analytics, caching, e-commerce

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KEY REMINDERS	
✔	Different databases suit different needs
✔	NoSQL = flexible, scalable, unstructured
✔	Cloud = managed, scalable, global
✔	Spatial = geographic, location-based
✔	In-Memory = ultra-fast, volatile
✗	There is no "one-size-fits-all" database

"The choice of database model depends on the data, the application, and the requirements—different problems require different solutions."